

Bachelor of Technology — Materials Science and Engineering

Table 1: Materials Science and Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	45
	Discipline Electives	21
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Materials Science and Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
2	MT-201	Physics of Solids		3
3	MT-202	Quantum Mechanics and Applications		3
4	MT-203	Material Synthesis and Characterisation		4
5	ME-100	Reverse Engineering	0-0-2-1	1
6	MT-204	Thermodynamics and Kinetics and Materials	3	3
7	MT-205	Functional Properties of Materials	4	4
8	MT-206	Extraction and Materials Processing	4	4
9	ME-212	Product Realization (Manufacturing) Technology	2-0-3-3	3
10	MT-301	Phase Transformations		3
11	MT-302	Transport Phenomena		3
12	MT-303	Computational Materials Science		4
13	MT-304	Mechanical Behaviour of Materials		4
14	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total				45

Table 3: B. Tech Materials Science and Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Materials Science and Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-241	Material Science for Engineers	3-0-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Materials Science and Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	DC	MT-201	Physics of Solids		3
5	DC	MT-202	Quantum Mechanics and Applications		3
6	DC	MT-203	Material Synthesis and Characterisation		4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Materials Science and Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-100	Reverse Engineering	0-0-2-1	1
2	DC	MT-204	Thermodynamics and Kinetics and Materials	3	3
3	DC	MT-205	Functional Properties of Materials	4	4
4	DC	MT-206	Extraction and Materials Processing	4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Materials Science and Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-212	Product Realization (Manufacturing) Technology	2-0-3-3	3
2	DC	MT-301	Phase Transformations		3
3	DC	MT-302	Transport Phenomena		3
4	DC	MT-303	Computational Materials Science		4
5	DC	MT-304	Mechanical Behaviour of Materials		4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Materials Science and Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Materials Science and Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Materials Science and Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MS-010	Internship		2
2	MTP	MS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Materials Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-303	Heat Transfer	3
2	ME-308	Manufacturing Engineering	3
3	ME-353	Electronic Materials and Their Applications	3
4	ME-515	Carbon Materials and Technology	3
5	MT-501	Energy Conversion and Storage Technologies	3
6	MT-502	Recycling and Circular Economy	3
7	MT-503	Semiconductor Materials and Devices	3
8	MT-504	Powder metallurgical processing of Materials	3
9	MT-505	Thin Film Technology	3
10	MT-506	Biomaterials	3

Materials Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	MT-507	Modelling and Simulations in Material Science	3
12	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
13	MS-498P	Major Technical Project - I (compulsory; counted as DE)	4
14	MS-499P	Major Technical Project - II (compulsory; counted as DE)	4